Computer Vision Project

Project Overview

You are required to build a modular **library of image processing functions and algorithms** that will be used to solve a set of practical computer vision problems.

Each team must design their own solution strategy and implement all required components from scratch using the concepts covered in the course and additional research when needed.

Important Instructions

- Number of students per team: **4–5 students**
- You are NOT provided with datasets or images.
- You are expected to search for and collect suitable images from public datasets or online sources.
- If any task description is unclear for you, you are encouraged to explore related documentation and research similar applications in the field of image processing and computer vision.
- You must design your own processing strategy for each task.
- Your implementation should be modular and reusable (library-style design).

Project Tasks

1. Selective Object Enhancement (Color-Based Editing Tool)

Design a system that enhances a specific object in an image while applying different visual effects to the rest of the image.

Example: In an image containing red roses and yellow flowers, the system should enhance (sharpen) only the red roses while applying a blurring effect to the yellow flowers (or vice versa).

2. Comparative Study of Sharpening Pipelines for X-Ray Images

Design and implement three different image enhancement strategies (pipelines) for enhancing X-ray image.

You are required to compare the results and analyze the strengths and weaknesses of each approach.

Note for you: similar to the enhancement pipeline studied in one of the laboratory sessions.

3. Intelligent Auto Image Enhancement System

Develop a system that automatically enhances an input image that may be dark, bright, or low contrast. The system must analyze the image characteristics and decide the most suitable enhancement strategy automatically.

4. Document Cleaning System

Design a system that enhances scanned document images affected by noise, shadows, uneven lighting, or degradation, producing a clean and readable output.

Reference: Use the same images provided in the following article:

https://medium.com/data-science/denoising-noisy-documents-6807c34730c4

5. Panorama Image Stitching

Develop a system that merges two overlapping images of the same scene into a single seamless panoramic image.

6. Transformed Object Recognition

Design a system capable of identifying an object in an image even if it has been rotated, translated, or slightly transformed.

7. Depth Approximation from Two Images

Develop a system that estimates relative depth information from two images of the same scene taken from slightly different viewpoints (e.g., two images captured from slightly different angles).

The goal is to analyze how objects shift between the two images and use this information to estimate which parts of the scene are closer or farther from the camera.

Your system should produce a visual representation (depth map) where:

- Brighter regions represent closer (larger shift)
- Darker regions represent farther objects (smaller shift)

8. High Dynamic Range (HDR) Imaging

Develop a system that combines multiple images of the same scene captured with different exposure levels into a single image with improved dynamic range and detail visibility.

You are required to implement the main components of the HDR process yourself. You are NOT allowed to rely on a single built-in function that performs the entire HDR pipeline automatically.

Submission Requirements

Each team must submit:

- A complete implementation of the image processing library.
- Source code for all tasks.

- A report explaining:
  - The design of your library
  - The approach used for each task
  - Results and observations
- Output images demonstrating each task.

Deadline

Friday, 8 May - 11:59 PM